

Title of thesis Second title line

Master Thesis of

Name

At the Department of Informatics and Mathematics
Chair of Theoretical Computer Science



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Statement of Authorship

I hereby declare that this document has been composed by myself and describes my own work, unless otherwise acknowledged in the text.

Passau, July 16, 2025

Abstract

A short summary of what is going on here.

Deutsche Zusammenfassung

Kurze Inhaltsangabe auf deutsch.

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1. Introduction

This chapter should contain

1. A short description of the thesis topic and its background.
2. An overview of related work in this field.
3. Contributions of the thesis.
4. Outline of the thesis.

2. Preliminaries

Before tackling the problem, we will introduce some concepts that are necessary for the understanding of this problem.

Definition 2.1 (Graph Embedding). *a Graph embedding of a graph G on a*

Definition 2.2 (Level Assignment). *A level assignment l of a level graph G is a mapping of vertices of G to levels.*

$$l : V(G) \rightarrow \mathbb{N}$$

Definition 2.3 (Proper Level Graph). *a Proper Level Graph G is a graph with a level assignment l that defines in which level every vertex is while respecting that for every edge $(u, v) \in E(G) \mid l(u) - l(v) \mid = 1$*

Definition 2.4 (Planar Graph). *a Planar Graph G is a graph that can have an embedding on a 2-dimensional Euclidean space $\mathcal{R}$ in a way that edges only intersect with each other at their endpoints.*

Definition 2.5 (2SAT).

3. Reduction of the equivalent classes

The content chapters of your thesis should of course be renamed. How many chapters you need to write depends on your thesis and cannot be said in general.

3.1. The induced Hannani-tutte from 2-SAT and planarity

...

3.2. Equivalent classes and Hannani-tutte

3.3. Reduction of the equivalent classes

After linking the Equivalent Classes to the hanani-tutte drawing induced by the 2-Sat formula (insert paper), and also presenting the cases that lead to a cyclic relation between some vertices of the same level, we can now present our algorithm that reduce the set equivalent classes resulted from resolving the 2-SAT formula (insert papers).

Definition 3.1 (level assignement). *level assignement $l : V \rightarrow \{1, \dots, k\}$ with $k \in \mathbb{N}$ that satisfies $l(u) + 1 = l(v)$ for each edge $(u, v) \in E$*

Definition 3.2. *adj outputs the distinct endpoints in a level h of adjacent edges with the shared source endpoint v from $h - 1$.*

$$\begin{aligned} \text{adj}: (l(V), V) &\rightarrow \mathcal{P}(V) \\ (h, v) &\mapsto \{w \mid w \in V, (v, w) \in E \text{ and } l(w) = h\} \end{aligned}$$

Definition 3.3. *the function of distinct endpoint of adjacent edges $\mathcal{V}_{\text{adj}}$*

$$\begin{aligned} \mathcal{V}_{\text{adj}}: (l(V), V) &\rightarrow \mathcal{P}(\mathcal{V}) \\ (h, v) &\mapsto \{wx \mid w, x \in \text{adj}(v, h) \text{ and } w.\text{id} < x.\text{id}\} \end{aligned}$$

Definition 3.4 (confused vertex). *A confused vertex v is a vertex that is not adjacent to an incoming edge from the level $l(v) - 1$.*

We define $\mathcal{V}_{\text{confused}}$ as follows

$$\begin{aligned} \mathcal{V}_{\text{confused}}: (l(V)) &\rightarrow \mathcal{P}(\mathcal{V}) \\ h &\mapsto \{v \mid v \in V, l(v) = h, \text{ and } \forall u \in V l(u) = l(v) - 1 \text{ and } (u, v) \notin E\} \end{aligned}$$

Based on the (insert section proof that we can fix the issue with adjacent edges by making them equal eachother) we can fix the problem of cyclic relations between adjacent edges, to specify an order we suggest going with the index order of vertices, to do so we will need to concatenate the the equivalent classe of vw and it's .. wv (since we respect the consistency clause) With the equivalent class that we got from the algorithm based on (insert paper),

3.3.1. Algorithm description

We have as input a planar proper level graph $G = (V, E, l)$ with a level assignement l , and equivalence classes based on the 2-SAT formulation of level planar embedding presented by Randert et al [RSB⁺01]. The 2-SAT formulation tests wether a proper level graph is planar. The truth assignements that solve the 2-SAT formula give embeddings that respect the planarity constraints but some of the assignments can't be translated to an embeddings in a plane surface due to the existence of cyclic relation between some of the vertices from the same level. These set of these assignements is equal to the set of possible combinations of assignement of the equivalence classes.

The goal of the algorithm is to merge some of the equivalence classes which means less possible truth assignement claiming that with the newly merged equivalence classes all the possible assignements of these equivalence classes lead to an embedding.

We first make the edges of G point to the same direction.(part of the computation of the equivalent classes)

We traverse the graph level by level, we also keep track of the vertices that we went throught by adding them to a graph G' that is initially an empty graph. Let h be the current level. In each level we go throught all the vertices, for each vertex v we execute two main taskes. The first task is to merge the equivalence classes of neighboring vertices to v that belong in the same adjacent level to level h . The second task is to merge the equivalence classes of pair of vertices (u, w) with u and w are in different connected components C_1 and C_2 of $G - \{v\}$ and v is a cut vertex of G connecting C_1 and C_2 .

For the first task, we consider vertices in $V_v^{(h-1)}$ adjacent to the vertex v on the level $h - 1$. We compute an ordering $\mathcal{V}$ of the vertices in $V_v^{(h-1)}$ so that for all pairs of vertices $(u, w) \in V_v^{(h-1)} \times V_v^{(h-1)}$ and $u < w$ the equivalence classes point to the same direction.

We merge those equivalence classes of the pairs $(u, w) \in \mathcal{V}$ with $u < w$. We merge also the equivalence classes of the pairs $(u, w) \in \mathcal{V}$ with $u > w$ to hold the consistency clause.

We will also do the same steps for vertices in $V_v^{(h+1)}$ that are adjacent to the vertex v on the level $h + 1$.

For the second task. we add v to G' . If the vertex v is a cut-vertex in G' , we look for the connected components S_c in $G' - \{v\}$. We pick an arbitrary absolute ordering $\mathcal{V}_{S_c}$ of the connected components in S_c . For each pair of connected components $(C_1, C_2) \in \mathcal{V}_{S_c}$, for each $w_1 \in C_1$ with $l(w_1) = h - 1$ we merge the equivalence classes of w_1 with all $w_2 \in C_2$ and $l(w_2) = h - 1$.

3.3.2. Correctness of the algorithm

The algorithm is based on the fact that we can convert a 2-SAT truth assignement to a strong hanani-tutte drawing and that we can convert a strong hanani-tutte drawing to a planar drawing.

The assignement of equivalent classes represent the set of solutions from solving the 2-SAT assignement.

The algorithm reduce the set of truth assignement solutions to get rid of false positives and that by reducing the number of equivalent classes by concatining some of them, which lead to smaller number of possible solution that can lead to a level planar drawing.

Proof. Since we are just reducing the set of truth assignement that already satisfies 2-SAT, the planarity and the consistency clauses are still respected.

if we take a truth assignement ϕ from 2-sat solutions, we can induce a strong hanani-tutte drawing. The goal is to convert this strong hanani-tutte drawing into a x-montone embedding which is a planar drawing.

To convert the strong hanani-tutte drawing to weak hanani-tutte, we must remove any odd crossing of adjacent edges by making it an even crossing. We either have two edges consecutive in the rotation of the shared endpoint, so we can just cross them one more time near the shared endpoint, this doesn't change the order of the vertices in the hanani-tutte drawing Figure 3.1. The other cases are when there is at least an edge between the two edges that crosses oddly and this edge crosses evenly all edges of the constructed graph so far, we have three major cases (TODO describing the cases from the hanani-tutte to xmonotone paper) , and we can get rid the odd crossing by concatinating the equivalent classes while still respecting the consistency clause Figure 3.2

After the modification, each element of the set of solutions from the reduced equivalent classes will lead to a weak hanani-tutte drawing, there is a special case in weak hanani-tutte that doesn't lead to an equivalent x-monotone embedding wich means that we need to change the relative order of some vertices. We mimick the construction in the proof of theorem 2.1 (reference hanani-tutte to xmononte paper) of the graph G' (which equals G at the end of the construction) , when the algorithm reach the special case which is when the new added x_1 adds more than one edge to the graph G' and it is not connected to the last added vertex before it in G' , from the proof we can notice that we fix the order of two connected comps to each other if they don't have any relative order (TODO explain what's a relative order in this context), if they do, meaning that one encapsulate the other then each vertex of the encapsulating component forces that it has the same equivalent class to all vertices of the encapsulated comp. From the algorithm we take an arbitrary order of the components (C_1, C_2) and we force all elements of C_2 to have the same equivalent class with the same vertex from C_1 . If C_1 is the encapsulated component, we just switch C_1 with C_2 .

□

Algorithm 3.1: EQUIVALENCE CLASSES REDUCTION

Input: proper level Graph $G = (V, E, l)$ and Eq
Output: Eq

```

1  $\mathcal{V} = \{uw \mid u, w \in V, u \neq w, l(u) = l(w)\}$ 
   /* reducing the equivalence classes of adjacent edges */
2 forall  $h \in l(V)$  do
   /* i still need to make sure adding  $h + 1$  is valid here. */
3   forall  $v \in \{V \mid l(v) \in \{h - 1, h + 1\}\}$  do
4      $S \leftarrow \emptyset$ 
5      $\mathcal{V}_{\text{ord}} \leftarrow \emptyset$ 
6     forall  $uw \in \mathcal{V}_{\text{adj}}(h, v)$  do
7       if  $wu \in S$  then
8          $\mathcal{V}_{\text{ord}} \leftarrow \mathcal{V}_{\text{ord}} \cup \{wu\}$ 
9          $S \leftarrow S \cup \text{Eq}(wu)$ 
10      else
11         $\mathcal{V}_{\text{ord}} \leftarrow \mathcal{V}_{\text{ord}} \cup \{uw\}$ 
12         $S \leftarrow S \cup \text{Eq}(uw)$ 
13     $\forall uw \in \mathcal{V}_{\text{ord}}: \text{Eq}(uw) = \bigcup_{u', w' \in \mathcal{V}_{\text{ord}}} \text{Eq}(u'w')$ 
   /* Synchronize with the symmetrical order to  $\mathcal{V}_{\text{ord}}$  */
14     $\forall uw \in \mathcal{V}_{\text{ord}}: \text{Eq}(wu) = \{xv, vx \in \text{Eq}(uw)\}$ 
   /* we have equivalent classes that can induce a weak
      hanani-tutte drawing */
15   forall  $v$  in vertices of level  $h$  do
16      $G' \rightarrow G' \cup \{(w, v) \mid w \in V, (w, v) \in E \text{ and } l(w) = l(v) + 1\}$ 
17     if  $v$  is a cut vertex of  $G'$  then
18       Comps list of connected comps of  $G' - \{v\}$ 
19       we pick an arbitrary order of the list Comps
20       forall  $(C_1, C_2)$  in combinaisons of pairs of connected components in
         Comps with the respect to the specified order do
21         if  $C_1$  is encapsulated by  $C_2$  then  $(C_1, C_2) \leftarrow (C_2, C_1)$ 
   /* we consider the vertices of the  $h$  level only */
22    $\forall v \in C_1(h) \forall w \in C_2(h) : \text{Eq}(v, w) \leftarrow \bigcup_{w' \in C_2(h)} \text{Eq}(v, w')$ 

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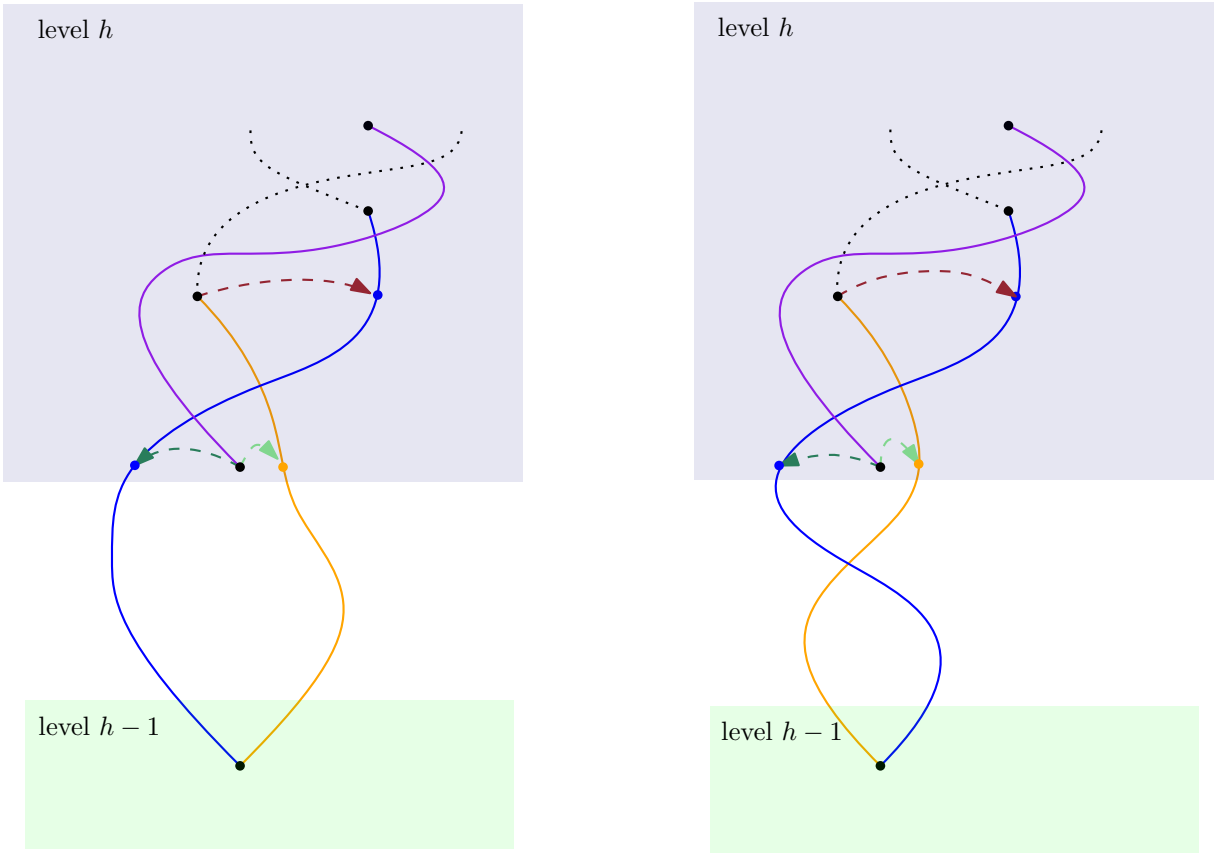


Figure 3.1.: consecutive edges in rotation that crosses oddly

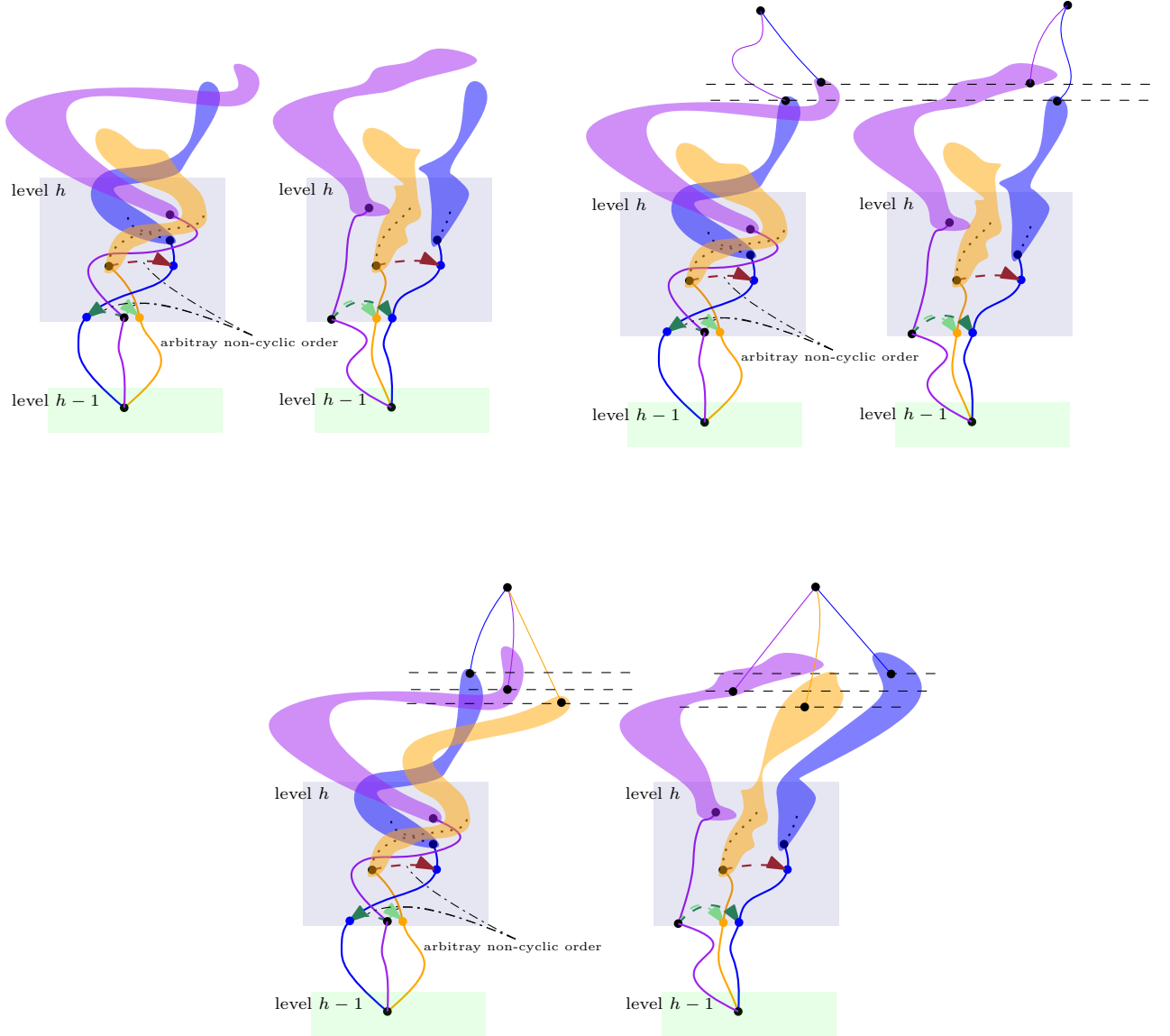


Figure 3.2.: in between even crossing edge

4. Conclusion

Summary and outlook.

Bibliography

- [RSB⁺01] Bert Randerath, Ewald Speckenmeyer, Endre Boros, Peter Hammer, Alex Kogan, Kazuhisa Makino, Bruno Simeone, and Ondrej Cepek. A satisfiability formulation of problems on level graphs. *Electronic Notes in Discrete Mathematics*, 9:269–277, 2001.

Appendix

A. Appendix Section 1

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Figure A.1.: A figure